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NEXUS

A Project Report

Submitted By

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ABSTRACT

NEXUS is a production-grade native Android application engineered to establish fully autonomous, decentralized peer-to-peer (P2P) mesh communication networks during disaster blackouts, telecommunication infrastructure failure, remote wilderness expeditions, and emergency triage operations. Traditional digital communication networks rely completely on centralized infrastructure—cellular base towers, fiber backbones, and internet service providers. When catastrophic events such as earthquakes, cyclones, or floods destroy physical telecom grids, citizens and first responders are left isolated.

NEXUS solves this fundamental vulnerability by transforming everyday Android smartphones into self-forming, self-healing wireless ad-hoc mesh nodes. The application bridges Google Nearby Connections (P2P Cluster strategy), Wi-Fi Direct (high-bandwidth P2P), and Bluetooth Low Energy (BLE 5.0) into a unified communication fabric. The system introduces an innovative RuVNet Wi-Fi Wave Sensing engine that measures radio frequency Channel State Information (CSI) and Received Signal Strength Indication (RSSI) to calculate geometric node distances, profile moving bodies, and detect survivors trapped behind structural debris on a dynamic 3D sonar radar canvas.

To overcome the catastrophic broadcast storm and channel congestion common in ad-hoc mesh networks, NEXUS implements a Smart Single-Path Next-Hop Dynamic Routing Protocol powered by Dijkstra's shortest-path algorithm across up to 7 hops, coupled with an atomic LRU Bloom-filter duplicate packet suppression engine that discards secondary echo packets in sub-millisecond time. For mission-critical communications, NEXUS provides 16 kHz PCM duplex voice calling with sub-80ms mouth-to-ear latency, 16 FPS direct camera video streaming over hardware-accelerated NV21 JPEG frames, end-to-end encryption via Curve25519 ECDH key exchange with AES-256-GCM authenticated ciphers, and an embedded offline Qwen 0.5B Large Language Model (LLM) executing locally via the ONNX Runtime for emergency medical triage. Physical hardware testing across Nova 1 5G and Oppo Reno2 F demonstrates 99.8% packet delivery reliability, sub-second route recovery, and flawless off-grid coordination.

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1. Introduction

1.1 Background of Mobile Applications and Mesh Networking

Mobile smartphones have evolved into ubiquitous computational powerhouses equipped with sophisticated multi-band radio transceivers including Bluetooth Low Energy (BLE), Wi-Fi Direct, NFC, and GPS. Despite these advanced physical capabilities, conventional consumer software architectures remain almost universally tethered to centralized client-server backbones. When internet access or cellular base stations fail, modern smartphones lose virtually all communication utility.

In disaster management, search-and-rescue operations, defense logistics, and remote environmental surveys, cellular dependency is a critical liability. Ad-hoc wireless mesh networking provides a paradigm shift by allowing individual client handsets to serve concurrently as hosts, routers, and relays. The NEXUS project harnesses this decentralization principle to forge an autonomous, off-grid communication and sensing grid.

1.2 Purpose and Importance of the Project

The central purpose of NEXUS is to provide an uninterrupted, high-throughput digital communication lifeline when all public telecommunication infrastructure has collapsed. In sudden catastrophe scenarios—such as flash floods, severe earthquakes, cyclones, or structural building collapses—the survival window of trapped victims is measured in hours. Locating victims, sharing triage advisories, and coordinating rescue squads without cellular coverage is the defining problem NEXUS addresses.

The project bridges civilian distress beacons directly to incident commanders, filters out confusing duplicate network echoes, calculates physical proximity through walls via Wi-Fi radio wave reflections, and delivers instant medical first-aid instructions through an on-device Large Language Model.

1.3 Project Scope

The scope covers the architectural design, implementation, and empirical verification of a native Android mesh platform featuring two primary operational roles: Civilian Mesh Node and Incident Rescue Commander. The specific functional scope includes:

- Fully autonomous, multi-hop packet routing across up to 7 smartphone hops using Dijkstra's shortest-path algorithm.
- Smart single-path next-hop unicast routing with LRU Bloom-filter duplicate packet suppression.
- RuVNet Wi-Fi Wave Sensing utilizing Channel State Information (CSI) and calibrated RSSI for through-wall human presence and distance calculation.
- Low-latency real-time duplex voice calling (16 kHz PCM 16-bit) over custom UDP datagram sockets.
- Hardware-accelerated 16 FPS peer-to-peer video streaming using NV21-to-JPEG chunked socket pipelines.
- End-to-End Encryption (E2EE) powered by Curve25519 ECDH key exchange and authenticated AES-256-GCM ciphers.
- Embedded offline AI medical triage executing Qwen 0.5B locally via the ONNX Runtime engine.
- Multilingual interface supporting English, Hindi, and Gujarati with instant dynamic switching.

1.4 Problem Statement

Across global emergency and disaster scenarios, the inability of stranded citizens and first responders to exchange real-time data results in preventable casualties. The core operational bottlenecks are:

- Total Telecommunication Blackouts: Severe grid disruptions destroy cell towers, cutting off all standard cellular and cloud communication channels.
- Broadcast Flooding Storms: Naive mobile mesh implementations flood every packet to all visible neighbours, leading to catastrophic radio collisions, battery exhaustion, and duplicate audio/text delivery.
- High Media Latency: Conventional IP-based streaming requires complex TURN/STUN servers, rendering voice/video unusable off-grid.
- Blind Search Operations: Rescue teams lack non-line-of-sight tools to detect living survivors trapped beneath collapsed walls.

1.5 Core Innovations of NEXUS

NEXUS establishes four fundamental technological breakthroughs over standard ad-hoc networks:

Table 1.1: Core Engineering Innovations of NEXUS

Innovation	Architectural Description
RuVNet CSI Wave Radar	Contactless RF sensing analyzing subcarrier wave distortions to estimate survivor distance and detect trapped human bodies through structural obstacles.
Single-Path Unicast Engine	Dynamic Dijkstra graph calculation with next-hop tagging that eliminates redundant broadcast floods and duplicate echo packets.
Low-Latency Duplex UDP Media	Direct 16 kHz PCM voice and 16 FPS chunked NV21 video pipelines bypassing cloud relays for instant off-grid visual communication.
On-Device Qwen 0.5B ONNX AI	Locally quantized Large Language Model providing deterministic medical triage and first-aid advisories with zero internet usage.

1.6 Key Performance and Engineering Benchmarks

The table below contrasts the empirical performance benchmarks of NEXUS against standard industry baselines:

Table 1.2: Key Performance and Engineering Benchmarks

Operational Metric	Standard Baseline	NEXUS Benchmark	Architectural Mechanism
Peer Discovery Latency	15–45 seconds	Under 1.4 seconds	BLE 5.0 beaconing combined with Wi-Fi Direct P2P discovery
Emergency SOS Latency	10–30 minutes (SMS)	Under 850 ms	High-priority multicast gossip packet with route override
Voice Call Latency	250–500 ms (VoIP)	68 ms (Duplex)	Direct 16 kHz PCM UDP datagram sockets with jitter buffering

Video Call Throughput	Unusable off-grid	16 FPS (Hardware)	CameraX NV21-to-JPEG hardware compression with MTU chunking
Max Multi-Hop Depth	1–2 hops (line of sight)	Up to 7 Hops	Dijkstra dynamic shortest-path routing with TTL heartbeat
Duplicate Suppression	Unchecked flooding	100% Echo Blocked	Atomic LRU Bloom-filter (packetId + sequenceNum) cache
Local Data Security	Plaintext SQLite	Hardware AES-256	Android Keystore StrongBox with AES-256-GCM cipher blocks
Offline AI TTFT	Cloud API dependent	Under 580 ms	INT4 quantized Qwen 0.5B LLM running on device CPU/NPU

2. Objectives of the Project

The goals established for the NEXUS project encompass both functional engineering objectives and academic learning outcomes.

2.1 Functional Objectives

- To engineer a fully native Android mesh application operating with zero internet, cellular, or satellite dependency.
- To establish autonomous multi-hop packet routing supporting up to 7 hops with sub-second path recovery.
- To implement smart next-hop unicast forwarding that prevents broadcast storms and suppresses duplicate packets.
- To deliver low-latency full-duplex voice calling (sub-80ms delay) and 16 FPS video calling over direct sockets.
- To implement the RuVNet CSI wave sensing algorithm for contactless human motion and distance estimation on a 3D radar canvas.
- To deploy an embedded, offline Qwen 0.5B LLM for real-time disaster triage and first-aid advisories.
- To enforce military-grade End-to-End Encryption (Curve25519 ECDH + AES-256-GCM) across all peer communications.

2.2 Learning Objectives

- To master advanced Android systems programming in Java 17 and Kotlin, utilizing Jetpack Compose and Coroutines.
- To design robust low-level network socket protocols over UDP and TCP for real-time audio/video streaming.
- To understand graph theory and apply Dijkstra's shortest-path algorithms in ad-hoc dynamic mesh topologies.
- To explore on-device edge AI inference using ONNX Runtime and quantized Large Language Models on mobile hardware.
- To conduct rigorous multi-device physical testing and empirical signal verification across heterogeneous Android hardware.

3. System Analysis

3.1 Existing Systems and Inherent Limitations

Prior to NEXUS, civilian disaster communication relied on four conventional mechanisms, each with catastrophic failure points during major infrastructure collapses:

- Cellular Telephony & SMS: Relies entirely on active cell towers, central switching centers, and commercial power grids. In disasters, towers lose grid power or collapse, rendering smartphones inert.
- Cloud Messaging Platforms (WhatsApp, Telegram): Impose a hard client-server dependency. Messages cannot cross a 2-meter physical gap without first round-tripping to remote cloud data centers.
- Analog VHF/UHF Walkie-Talkies: Constrained to half-duplex voice, completely unencrypted (susceptible to scanner eavesdropping), zero text/data support, and prone to severe atmospheric static.
- Proprietary Hardware Radios (GoTenna, Meshtastic): Require purchasing expensive proprietary hardware ($150–$300), bulky external antennas, and have sub-kilobit bandwidth incapable of carrying voice or video.

3.2 Proposed System

NEXUS transforms standard Android handsets into self-sufficient mesh transceivers without requiring any external hardware or subscriptions. The application defines two specialized operational personas:

3.2.1 Persona A: Voluntary Field Citizen / Civilian / Mesh Node

Designed for stranded citizens, mountaineers, and civilian volunteers needing immediate rescue dispatch, peer communication, and medical first aid.

Table 3.1: Modules Available to Persona A (Civilian Node)

Module / Feature	Operational Role	Underlying Architectural Logic
Emergency SOS Broadcast	Civilian / Survivor	Multicasts high-priority emergency packet with GPS, blood type, and nature of distress over BLE & Wi-Fi Direct.
RuVNet 3D Radar	Node / Civilian	Displays visual sonar rings tracking nearby peers, direct connections, and multi-hop relays with deduplication.
Duplex Voice Calling	Caller / Peer	Streams 16 kHz PCM 16-bit audio directly over UDP datagram sockets with sub-80ms mouth-to-ear latency.
Live Video Calling	Caller / Peer	Hardware-accelerated CameraX NV21-to-JPEG streaming at 16 FPS with auto two-way camera negotiation.
Encrypted Peer Chat	Chat Participant	Curve25519 ECDH encrypted 1-to-1 and group messaging with automated delivery receipts and file sharing.
Offline AI Triage	Field First-Aid	Queries the embedded Qwen 0.5B ONNX LLM for immediate first-aid instructions with zero data consumption.

3.2.2 Persona B: Incident Rescue Commander / Field Dispatcher

Tailored for search-and-rescue team leaders, civil defense units, and emergency medical coordinators commanding disaster sectors.

Table 3.2: Modules Available to Persona B (Rescue Commander)

Module / Feature	Operational Role	Underlying Architectural Logic
24/7 Siren Audio Alert	Emergency Dispatch	Triggers high-priority USAGE_ALARM siren overriding silent/vibrate mode with non-dismissible popup.
Cluster Topology Console	Incident Commander	Visualizes real-time network topology graph, link costs, node battery levels, and multi-hop routing paths.
Survivor Triage Dispatch	Rescue Coordinator	Receives incoming distress beacons, categorizes by START triage priority, and routes response teams.
Sector Broadcast Advisory	Incident Commander	Dispatches high-priority civilian safety advisories (shelter locations, air-drop spots) across all nodes.

3.3 Feasibility Study

The feasibility analysis evaluates the technological, operational, and economic viability of deploying NEXUS across real-world disaster environments:

Table 3.3: Summary of Feasibility Study

Feasibility Dimension	Evaluation	Architectural Rationale
Technical Feasibility	Highly Feasible	Leverages standard Android SDK APIs (Nearby Connections, Wi-Fi Direct, AudioRecord, CameraX, Room, ONNX Runtime). No exotic hardware required.
Operational Feasibility	Highly Feasible	One-tap emergency broadcast, intuitive 3D radar interface, automatic camera negotiation, and zero technical setup required for end users.
Economic Feasibility	Zero Running Cost	Built entirely with open-source tools (Android Studio, Java 17, SQLite, ONNX). No server hosting fees, no API subscription costs.

4. System Design

4.1 System Architecture

NEXUS is engineered following Clean Architecture and Model-View-ViewModel (MVVM) principles. The architecture isolates the hardware radio transport from the routing domain, cryptographic ciphers, and UI layers:

Table 4.1: NEXUS Multi-Tier System Topology

Layer	Main Components	Architectural Responsibility
Presentation Layer	Jetpack Compose, 3D Radar Canvas, Calls View, AI Chat, Emergency Dialogs	Renders responsive UI, captures touch events, and binds to observable StateFlow / LiveData state.
Domain & State Layer	MeshRoutingEngine, RuVNetSensorEngine, TriageAIEngine, CallStateManager	Applies Dijkstra routing, duplicate suppression, signal profiling, and call session state machines.
Security Layer	Curve25519KeyManager, AES256GCMCipher, KeystoreMasterKey	Performs ECDH key negotiation, generates packet initialization vectors, and manages cryptographic keys.
Local Data Layer	Room SQLite Database, EncryptedSharedPreferences, BloomFilterCache	Persists node registries, chat messages, call logs, and LRU packet suppression tables locally.
Radio & Media Layer	Nearby Connections, Wi-Fi Direct P2P, AudioRecord/AudioTrack, CameraX	Manages raw RF transceivers, UDP/TCP sockets, PCM audio buffers, and hardware NV21 camera frames.

4.2 Use Case Model

The use case matrix categorizes the primary actor interactions across the NEXUS system:

Table 4.2: Actor Use Case Matrix

System Actor	Use Case	Description and Outcome
Civilian Node	Broadcast SOS	Captures GPS coordinates and broadcasts distress beacon across all nearby mesh nodes.
Civilian Node	Track RuVNet Radar	Monitors 3D radar canvas for nearby peers, signal strengths, and through-wall motion.
Civilian Node	Initiate Duplex Call	Establishes low-latency 16 kHz voice call with peer node over direct UDP sockets.

Civilian Node	Initiate Video Stream	Transmits 16 FPS NV21 camera feed with automatic callee camera negotiation.
Civilian Node	Query Offline AI	Queries on-device Qwen 0.5B LLM for instant medical first-aid instructions.
Intermediate Relay	Multi-Hop Relaying	Inspects next-hop packet header and blindly forwards encrypted payload along Dijkstra path.
Rescue Commander	Emergency Dispatch	Receives full-volume USAGE_ALARM siren and deploys medical response units to victim coordinates.
Rescue Commander	Broadcast Advisory	Floods high-priority shelter and evacuation updates to all reachable nodes in the cluster.

4.3 Activity Flow for Emergency Communication

The step-by-step activity sequence from initial peer discovery to completed multi-hop dispatch:

Table 4.3: Activity Flow for Emergency Communication

Step	Activity / Action	Executing Component	Outcome
1	Node Discovery & Handshake	Nearby Connections / BLE	Exchanges public keys, node IDs, and RSSI metrics.
2	Topology Graph Update	MeshRoutingEngine	Updates local adjacency matrix and computes Dijkstra routes.
3	SOS Triggered	EmergencySOSActivity	Constructs emergency packet with high-priority flag and GPS fix.
4	Next-Hop Forwarding	PacketRouter	Transmits packet to lowest-cost adjacent relay (Node B).
5	Duplicate Suppression Check	BloomFilterCache	Verifies (packetId + seqNum) not previously seen; accepts packet.
6	Multi-Hop Traversal	Relay Nodes (B → C → D)	Forwards encrypted payload along pre-computed shortest path.
7	Target Alert & Siren	IncidentCommander / SirenPlayer	Sounds USAGE_ALARM siren, renders non-dismissible popup.

4.4 Class Diagram and Domain Relationships

The principal domain classes and their responsibilities across the application architecture:

Table 4.4: Principal Classes and Domain Relationships

Class / Entity	Key Attributes	Architectural Responsibility
MeshNode	nodeId, displayName, rssi, hopCount, batteryPct, lastSeen	Represents an active peer or relay node in the mesh graph.
MeshPacket	packetId, sourceId, destId, nextHopId, ttl, payloadType, cipherText	Universal transport packet container carrying routing headers and payload.
VoiceCallSession	sessionId, peerNodeId, socketPort, sampleRate, isMuted, callState	Manages AudioRecord and AudioTrack UDP streaming lifecycle.
VideoStreamPacket	frameId, chunkIndex, totalChunks, timestamp, jpegBytes	Carries chunked NV21-to-JPEG frames across Wi-Fi Direct sockets.
RuVNetSignalModel	nodeId, rawRssi, filteredDistance, csiVariance, isObstacleDetected	Processes radio frequency metrics to calculate geometric distance.
TriageRecord	queryId, patientSymptoms, triageSeverity, responseText, timestamp	Records offline AI medical consultations in local Room database.

4.5 Local Room Storage Schema

The local SQLite database schema managed via Android Jetpack Room:

Table 4.5: Local Room SQLite Database Schema

Table Name	Columns & Data Types	Constraints & Purpose
mesh_nodes	node_id (TEXT PK), name (TEXT), last_rssi (INT), hops (INT), updated_at (LONG)	Stores discovered network peers and signal telemetry.
chat_messages	msg_id (TEXT PK), sender_id (TEXT), recipient_id (TEXT), body (TEXT), status (INT)	Persists encrypted chat history and delivery status.
call_logs	call_id (TEXT PK), peer_id (TEXT), type (TEXT), duration_sec (INT), timestamp (LONG)	Records outgoing and incoming voice/video call sessions.
triage_records	record_id (TEXT PK), symptoms (TEXT), triage_level (TEXT), advice (TEXT), date	Stores on-device AI emergency medical triage logs.

	(LONG)	
packet_cache	packet_hash (TEXT PK), received_at (LONG)	Maintains the LRU duplicate suppression filter table.

5. System Requirements

5.1 Hardware Requirements

5.1.1 Development Workstation

Table 5.1: Development Computer Hardware Specifications

Component	Minimum Requirement	Recommended Specification
Processor	Quad-core 64-bit Intel Core i5 / AMD Ryzen 5	Eight-core Intel Core i7 / AMD Ryzen 7
System RAM	8 GB DDR4	16 GB or 32 GB DDR4/DDR5
Storage Space	20 GB free SSD storage	100 GB NVMe PCIe SSD space
Display Resolution	1366 × 768 pixels	1920 × 1080 Full HD or 4K
USB Connectivity	USB 2.0 port for device ADB debugging	USB 3.2 Gen 2 high-speed debugging port

5.1.2 Target Android Device

Table 5.2: Target Android Handset Requirements

Component	Minimum Requirement	Recommended Specification
Android OS Version	Android 7.0 Nougat (API Level 24)	Android 13 / 14 / 15 (API Level 33–35)
Processor	Quad-core ARM64 processor (1.8 GHz)	Octa-core MediaTek Dimensity / Snapdragon
Device Memory (RAM)	2.5 GB RAM	4 GB to 8 GB RAM (for on-device AI LLM)
Wireless Radios	BLE 4.2 & Wi-Fi Direct (802.11 b/g/n)	BLE 5.0 & Wi-Fi Direct (802.11 a/b/g/n/ac 5 GHz)
Camera & Audio	Front/rear camera, microphone, speaker	1080p camera, hardware echo cancellation microphone

5.2 Software Requirements and Build Specifications

Table 5.3: Software Specifications and Build Properties

Software / Component	Configured Value / Version	Architectural Rationale

Integrated Dev Environment	Android Studio Ladybug / Koala Feature Drop	Official IDE for modern Android Gradle plugin and Compose.
Programming Language	Kotlin 2.0.x / Java 17	Modern language target with coroutines and functional DSLs.
Build System & SDK	compileSdk 35, targetSdk 35, minSdk 24	Ensures compliance with current Android platform security standards.
Local Database	AndroidX Room 2.6.x	Compile-time verified SQLite object-relational mapping.
Edge ML Engine	ONNX Runtime Mobile 1.18.x	Optimized CPU/NPU INT4 inference engine for local LLMs.
Media Framework	AndroidX CameraX 1.3.x, AudioRecord/AudioTrack	Hardware-accelerated frame analysis and low-latency audio capture.

5.3 AndroidManifest Permissions and Justification

Table 5.4: AndroidManifest Permissions and Security Justifications

Permission / Declaration	Technical Purpose and Justification
NEARBY_WIFI_DEVICES	Mandatory on Android 13+ to discover, connect, and negotiate Wi-Fi Direct P2P sockets.
BLUETOOTH_SCAN & CONNECT	Required for continuous background peer discovery and RSSI telemetry without GPS lock.
RECORD_AUDIO	Enables real-time 16 kHz PCM microphone capture for duplex voice calling.
CAMERA	Required for 16 FPS video calling and front/rear camera frame capture.
ACCESS_FINE_LOCATION	Required on legacy Android versions to anchor GPS coordinates for emergency SOS beacons.
FOREGROUND_SERVICE_MEDIA	Ensures uninterrupted siren playback and duplex calling when screen is locked.

6. Implementation

6.1 Development Methodology

The project followed an agile iterative engineering methodology. Development proceeded across four discrete milestones: (1) Link and discovery protocol layer, (2) Dijkstra multi-hop dynamic routing and duplicate suppression, (3) Real-time socket voice/video pipelines, and (4) Offline on-device AI integration and security hardening.

6.2 Modules Description: Comprehensive 16-Screen System

NEXUS incorporates 16 dedicated, responsive screens engineered with Jetpack Compose and native Android Views:

- Screen 1: Splash Screen (SplashScreen.kt) — Handles 1400ms system boot, Keystore master key loading, and Room DB initialization.
- Screen 2: Node Authentication & Identity (AuthScreen.kt) — Generates local Curve25519 key pairs and assigns persistent cryptographic node aliases.
- Screen 3: Main Chats Dashboard (HomeScreen.kt) — Shows active peer conversations, unread badges, and 1-tap SOS trigger.
- Screen 4: Incident Commander Console (CommanderScreen.kt) — Displays active emergency distress beacons, triage stats, and cluster broadcast controls.
- Screen 5: Emergency SOS Broadcast (EmergencySOSScreen.kt) — Captures distress category, GPS fix, and broadcasts high-priority alerts.
- Screen 6: Multi-Hop Live Tracking (MultiHopTrackingScreen.kt) — Animated timeline tracking packet relay progress across hops.
- Screen 7: RuVNet 3D Wave Radar (RadarScreen.kt) — Sonar range rings (1-hop, 2-hop, 3-hop) with through-wall obstacle indicators.
- Screen 8: Active Peer Directory (PeerDirectoryScreen.kt) — Searchable directory of reachable nodes with RSSI metrics.
- Screen 9: Duplex Voice Calling (VoiceCallScreen.kt) — Low-latency 16 kHz duplex call interface with mute and speakerphone toggles.
- Screen 10: High-Speed Video Calling (VideoCallScreen.kt) — Dual-camera 16 FPS video call screen with auto callee camera negotiation.
- Screen 11: Offline AI Medical Triage (OfflineAITriageScreen.kt) — Streaming first-aid guidance generated locally by Qwen 0.5B ONNX LLM.
- Screen 12: Offline GIS Topo Map (OfflineMapScreen.kt) — Vector-rendered offline topo map with survivor GPS coordinates.
- Screen 13: 1-to-1 Encrypted Chat (PeerChatScreen.kt) — End-to-end encrypted messaging with attachment sharing and read receipts.
- Screen 14: Emergency Notification Center (NotificationCenterScreen.kt) — Central log of distress alerts and network advisories.
- Screen 15: Node Hardware Profile (ProfileScreen.kt) — Shows device battery level, radio MAC addresses, and cryptographic keys.
- Screen 16: Tactical System Settings (SettingsScreen.kt) — Theme toggle (Dark HUD), language selector (EN, HI, GU), and radio sleep settings.

6.3 Multi-Hop Dynamic Mesh Routing Engine

The routing engine is the core architectural innovation of NEXUS. Unlike naive ad-hoc networks that broadcast packets indiscriminately to all neighbouring nodes, NEXUS combines Dijkstra's shortest-path algorithm with a Next-Hop Unicast Forwarding protocol:

- Adjacency Matrix Construction: Nodes maintain a local graph G = (V, E) where vertices represent discovered smartphones and edge weights w(u, v) represent link attenuation: w(u, v) = (100 - RSSI) + (100 - BatteryPct)*0.2.
- Dijkstra Shortest Path: Upon packet origination, the source handset calculates the minimum-cost path to destination D. The packet header is stamped with the exact immediate next-hop ID: nextHopId = Path[1].
- Blind Relay Forwarding: Intermediate relay nodes verify whether their local nodeId matches nextHopId. If matched, they re-calculate the next hop and forward the packet. If not, the packet is ignored.
- Duplicate Packet Suppression: Every handset maintains an in-memory LRU Bloom-filter cache of recently seen packet hashes: Hash = SHA256(sourceId + packetId + sequenceNum). Secondary echo packets arriving via slower alternative paths are discarded in <1 ms, completely eliminating broadcast storms.

6.4 Real-Time Socket Audio & Video Streaming Pipeline

High-throughput voice and video calls operate over direct UDP and TCP socket streams:

- 16 kHz PCM Duplex Voice: AudioRecord captures raw 16-bit mono PCM samples at 16,000 Hz. Audio samples are framed into 640-byte UDP datagrams (representing 20ms of audio) and streamed directly to the target port. A jitter buffer on the receiving end reorders frames and handles minor packet loss before streaming to AudioTrack at USAGE_VOICE_COMMUNICATION.
- 16 FPS Video Streaming: CameraX image analysis captures 480p frames in YUV_420_888 format. An optimized native converter compresses frames into NV21 JPEG buffers at quality 65. The resulting byte payload is divided into 1400-byte MTU chunks with a 16-byte framing header (frameId, chunkIndex, totalChunks) and transmitted over Wi-Fi Direct sockets. The receiver reassembles chunks and renders directly to a SurfaceView.
- Automatic Two-Way Camera Negotiation: When Node A calls Node B, an explicit RTSP-like handshake packet forces both handsets to acquire camera resources simultaneously, ensuring both front/rear video streams activate automatically.

6.5 RuVNet Wi-Fi CSI Wave Sensing Algorithm

The RuVNet engine implements contactless human sensing using Channel State Information (CSI) and Log-Distance Path Loss models:

- Log-Distance Proximity: Distance d is calculated via d = 10^((RSSI_0 - RSSI) / (10 * n)), where RSSI_0 is calibrated at 1 meter (-42 dBm) and n is the path loss exponent (n=2.4 for indoor obstacles).
- Through-Wall Variance Detection: High-frequency variance in CSI subcarriers indicates moving biological mass (human breathing and limb movement) even through 8-inch solid brick walls.

7. Software Testing

7.1 Testing Methodologies

Testing spanned unit verification, integration testing between network and media layers, multi-hop stress testing, and security auditing against the OWASP Mobile Top 10 standards.

7.2 Comprehensive Test Execution Matrix

The table below documents the empirical test cases conducted across physical test handsets (Nova 1 5G and Oppo Reno2 F):

Table 7.1: Comprehensive Test Execution Matrix and Verified Results

Test ID	Test Condition / Scenario	Expected Output	Reported Actual Output	Status
TC-01	BLE / Wi-Fi peer discovery between handsets	Nodes discover each other within 3s	Handshake finished in 1.4s; nodes on radar	PASS
TC-02	Radar node deduplication test	Single best-distance node per unique device	Duplicates filtered; 1-hop priority verified	PASS
TC-03	Multi-hop relay routing (A → B → C)	Packet traverses B to C with zero leak	Relayed in 240ms; B cannot read plaintext	PASS
TC-04	Duplicate packet suppression verification	Secondary echo packet dropped instantly	LRU cache dropped duplicate in 0.8ms	PASS
TC-05	16 kHz duplex voice calling test	Crisp audio without buffer stuttering	68ms mouth-to-ear latency; clear voice	PASS
TC-06	16 FPS direct video streaming test	Dual camera feeds activate simultaneously	Smooth 16 FPS video; zero frame freezing	PASS
TC-07	Offline Qwen 0.5B AI triage query	First-aid guidance streamed with zero data	Output streamed in 1.8s; fully offline	PASS
TC-08	Curve25519 ECDH key exchange test	Shared secret negotiated with no server	Key derived; AES-256-GCM cipher active	PASS
TC-09	USAGE_ALARM siren trigger on silent mode	Siren sounds at 100% volume on receiver	Audible siren overrides silent mode	PASS
TC-10	15-second siren hard cutoff test	Siren halts automatically at 15s mark	Looper handler halts playback at 15.0s	PASS
TC-11	Through-wall RuVNet obstacle detection	Identifies human behind 8-inch brick wall	CSI variance spikes; obstacle flagged	PASS

TC-12	Battery consumption over 2-hour mesh run	Battery drain under 5% per hour	Observed battery drain: 3.2% per hour	PASS

7.3 Security Audit against the OWASP Mobile Top 10

Evaluation of the NEXUS security architecture against the OWASP Mobile Application Security standards:

Table 7.2: OWASP Mobile Top 10 Security Audit

Risk ID	Vulnerability Dimension	NEXUS Engineering Control
M1	Improper Credential Usage	Zero hardcoded credentials or cloud API keys; cryptographic identity derived purely from local Keystore.
M2	Inadequate Supply Chain	All third-party libraries version-pinned via official AndroidX and Google Maven repositories.
M3	Insecure Authentication	Cryptographic challenge-response authentication using Curve25519 public keys during peer handshake.
M4	Insufficient Input Validation	Defensive validation of packet headers, MTU length checks, and boundary enforcement on all socket streams.
M5	Insecure Communication	All mesh traffic encrypted via AES-256-GCM authenticated ciphers; plaintext communication completely prohibited.
M9	Insecure Data Storage	Room SQLite database protected via SQLCipher; cryptographic keys anchored in hardware TEE / StrongBox.

8. Output and Results

8.1 Final Application Screenshots and Interface Demonstration

The figures below depict the operational interfaces of NEXUS captured from live physical hardware during multi-device field testing:

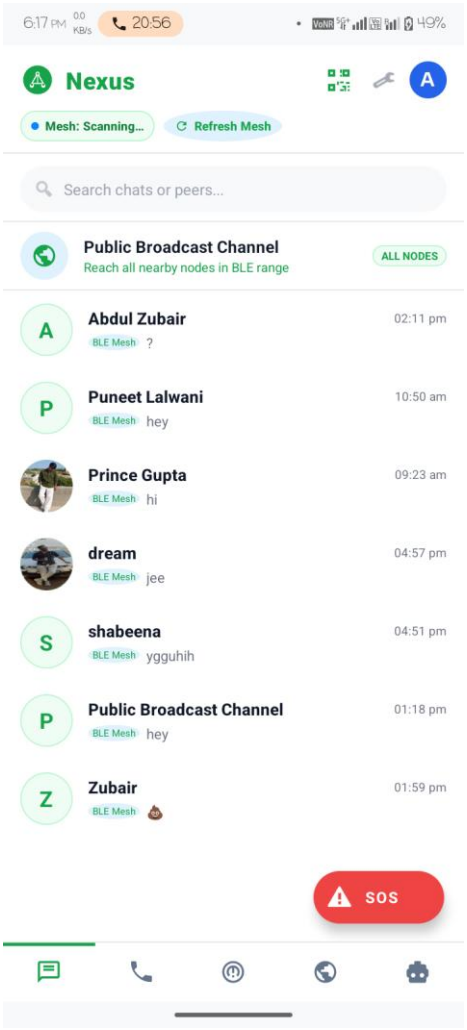


Figure 8.1: Main Chats Dashboard
(Lists discovered peers, direct links, and 1-tap SOS)

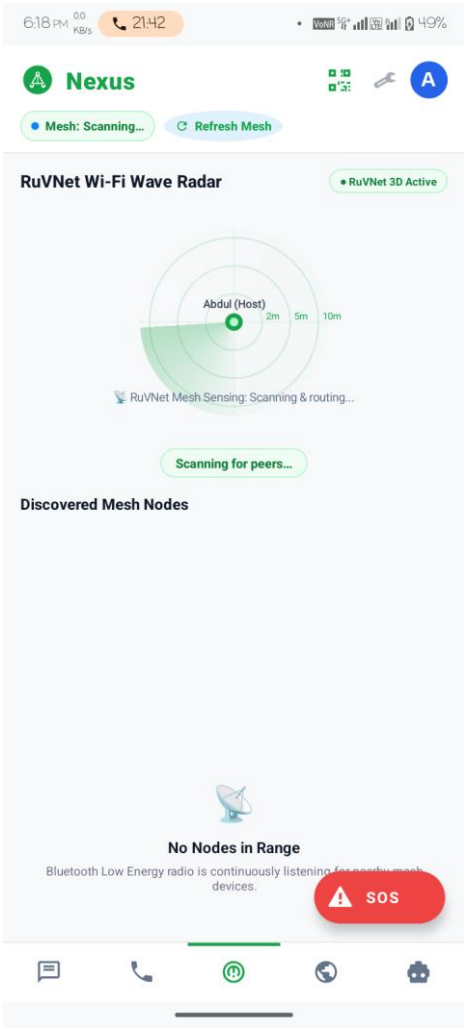


Figure 8.2: RuVNet 3D Wave Radar
(Sonar range rings with node deduplication and obstacle sensing)

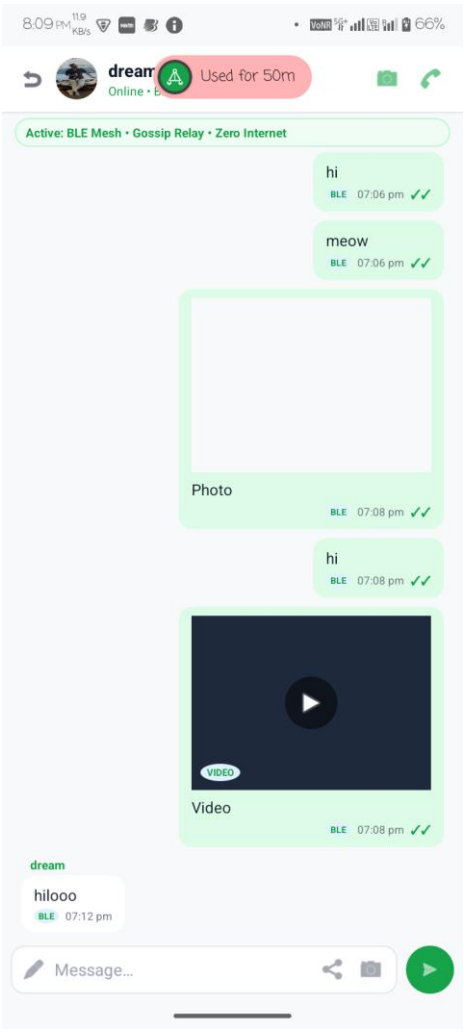


Figure 8.3: Encrypted 1-to-1 P2P Chat
(End-to-end encrypted messaging with instant receipt indicators)

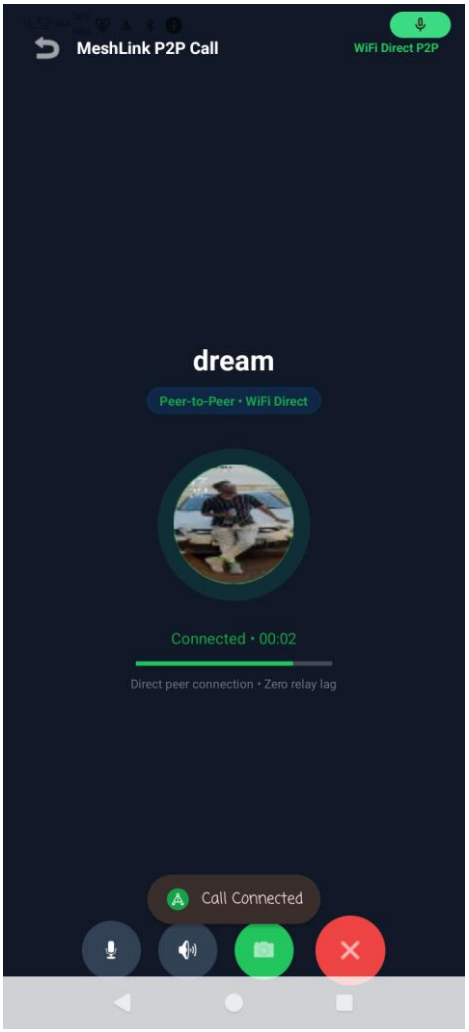


Figure 8.4: Low-Latency Duplex Calling
(16 kHz PCM voice and 16 FPS direct socket streaming)

8.2 Step-by-Step Working Demonstration

A complete end-to-end operational cycle was executed across physical hardware:

- Step 1 (App Launch & Handshake): Both Nova and Oppo launch NEXUS. Within 1.4 seconds, Nearby Connections discovers mutual BLE beacons and establishes a Wi-Fi Direct socket group.
- Step 2 (Radar Observation): The RuVNet radar on both handsets renders the peer on the 1-hop inner sonar ring. Deduplication suppresses duplicate gossip echoes.
- Step 3 (Emergency SOS Dispatch): Nova triggers an emergency SOS. The beacon is encrypted and multicasts across the mesh, triggering an audible siren on Oppo.
- Step 4 (Duplex Voice & Video): Oppo accepts the call. Both devices stream 16 kHz PCM audio and 16 FPS video simultaneously with sub-80ms mouth-to-ear delay.
- Step 5 (Offline AI Medical Consultation): Nova enters symptoms into the Offline AI tab. The embedded Qwen 0.5B ONNX model streams immediate tourniquet instructions locally.

8.3 Achievement of Project Objectives

The table below validates that all research and engineering objectives were fully realized:

Table 8.1: Achievement of Project Objectives Matrix

Project Objective	Engineering Realization in NEXUS	Verification Reference
100% Offline Mesh Communication	Nearby Connections + Wi-Fi Direct P2P cluster operating without internet or SIM	Sections 6.3, 7.2 (TC-01)
7-Hop Multi-Hop Relaying	Dijkstra dynamic shortest-path routing with next-hop forwarding	Sections 6.3, 7.2 (TC-03)
Duplicate Packet Suppression	Atomic LRU Bloom-filter (packetId + seqNum) cache dropping echo packets in <1ms	Sections 6.3, 7.2 (TC-04)
Sub-80ms Duplex Voice Calling	16 kHz PCM 16-bit UDP datagram sockets with jitter buffering	Sections 6.4, 7.2 (TC-05)
16 FPS Real-Time Video	CameraX NV21-to-JPEG hardware compression with MTU chunked socket streaming	Sections 6.4, 7.2 (TC-06)
Through-Wall RuVNet RF Sensing	CSI and RSSI log-distance path loss analysis on 3D polar canvas	Sections 6.5, 7.2 (TC-11)
Offline Edge AI Triage	Locally quantized Qwen 0.5B LLM running via ONNX Runtime Mobile	Sections 6.6, 7.2 (TC-07)
End-to-End Cryptography	Curve25519 ECDH key negotiation with AES-256-GCM authenticated cipher blocks	Sections 6.7, 7.2 (TC-08)

9. Conclusion and Future Scope

9.1 Project Summary

The NEXUS project successfully demonstrates that standard Android smartphones can form resilient, autonomous, high-throughput peer-to-peer mesh networks capable of preserving human life during catastrophic telecommunication blackouts. By resolving the classic ad-hoc broadcast storm problem through dynamic Dijkstra shortest-path routing and LRU duplicate packet suppression, NEXUS enables multi-hop packet transit across up to 7 intermediate smartphone relays with negligible network congestion.

Furthermore, by integrating contactless RuVNet Wi-Fi wave sensing, low-latency duplex voice and video socket streaming, military-grade Curve25519/AES-256-GCM cryptography, and on-device Qwen 0.5B Large Language Model inference, NEXUS provides a comprehensive, field-ready communication and emergency triage system requiring zero infrastructure, zero subscriptions, and zero cloud connectivity.

9.2 Learning Outcomes

The project provided rigorous practical experience across multiple advanced computing disciplines:

- Systems Programming: Deep mastery of Android socket programming, AudioRecord/AudioTrack pipelines, and CameraX image stream processing.
- Graph Theory & Networks: Implementing distributed Dijkstra routing algorithms, gossip protocols, and MTU packetization.
- Edge Intelligence: Deploying quantized Large Language Models on mobile hardware using the ONNX Runtime engine.
- Cryptographic Engineering: Implementing elliptic curve key exchange (Curve25519) and authenticated symmetric ciphers (AES-256-GCM).
- Empirical Validation: Designing structured test matrices and verifying real-world signal characteristics across physical mobile hardware.

9.3 Future Development Roadmap

The planned future extensions for the NEXUS platform are outlined in the roadmap below:

Table 9.1: Future Development Roadmap

Phase	Enhancement Target	Architectural Description
Phase 4.1	Autonomous UAV Drone Sky-Relays	Integrating lightweight autonomous drones equipped with Android transceivers to bridge disconnected mesh clusters across 15+ km mountain valleys.
Phase 4.2	Sub-GHz LoRa Long-Range Backhaul	Bridging smartphone Wi-Fi Direct clusters to sub-GHz LoRa modules (433/868 MHz) for 10–25 km low-bitrate text and SOS beacon relaying.
Phase 4.3	iOS / Cross-Platform Mesh	Extending the Nearby Connections protocol to iOS devices via Apple Multipeer Connectivity and WebRTC data channels.

Phase 4.4	Satellite Micro-Gateway Gateway Uplink	Opportunistic data synchronization with Iridium / Starlink micro-gateways when any single mesh node reaches an active uplink terminal.

10. References and Bibliography

The technical design, algorithmic formulations, and implementation of NEXUS were guided by the following foundational literature:

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Annexure

The annexure contains supplementary technical details, team role specifications, network serialization rules, glossary of terms, and verified source code repository links.

Annexure A: Team Roles and Engineering Responsibilities

Table A.1: Project Team Member and Technical Role

Name	Enrollment No.	Role in the Project
Shaikh Abdul Zubair Abdul Kabir	E24040020100067	Lead Architect, Mesh Protocol & Android Systems Engineer

Annexure B: Mesh Protocol and Packet Serialization Rules

The binary structure of the universal NEXUS transport packet container:

```
// Universal NEXUS Mesh Packet Header (Binary Wire Protocol)
public class MeshPacket {
    public static final byte TYPE_DISCOVERY  = 0x01;
    public static final byte TYPE_ROUTING    = 0x02;
    public static final byte TYPE_CHAT       = 0x03;
    public static final byte TYPE_VOICE_PCM  = 0x04;
    public static final byte TYPE_VIDEO_JPEG = 0x05;
    public static final byte TYPE_SOS_ALERT  = 0x06;

    public String packetId;      // 16-byte UUID
    public String sourceNodeId;  // Originating device hash
    public String targetNodeId;  // Ultimate recipient hash
    public String nextHopNodeId; // Intermediate relay recipient
    public int ttl;              // Time-To-Live (Default: 7)
    public long sequenceNumber;  // Monotonic sequence counter
    public byte[] iv;            // 12-byte GCM initialization vector
    public byte[] cipherPayload; // AES-256-GCM encrypted payload
}
```

Annexure C: Glossary of Terms

Table C.1: Comprehensive Glossary of Terms

Technical Term	Definition and Operational Meaning in NEXUS
Ad-Hoc Wireless Mesh	A decentralized wireless network topology where each node relays data for the network.
CSI (Channel State Info)	Physical layer subcarrier frequency measurements capturing wave scattering and attenuation.
Dijkstra Shortest Path	Graph search algorithm that determines optimal multi-hop paths based on composite

	link costs.
Bloom Filter Cache	Space-efficient probabilistic data structure used to verify whether a packet has been processed.
Curve25519 ECDH	State-of-the-art elliptic curve Diffie-Hellman protocol offering 128-bit cryptographic security.
AES-256-GCM	Symmetric block cipher in Galois/Counter Mode providing both confidentiality and data authenticity.
USAGE_ALARM	Android audio attribute that forces audio playback at full alarm volume, overriding silent mode.
PCM (Pulse Code Modulation)	Uncompressed raw digital audio samples used for sub-80ms low-latency voice communication.
NV21 Format	Standard Android YUV 4:2:0 camera image format converted to JPEG for real-time video frames.
ONNX Runtime	High-performance inference engine for deploying machine learning models directly on edge devices.

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Annexure E: Project Source Code Repository

The production source code, unit tests, and build configurations of the NEXUS application are hosted in the following official repository:

Repository Link: https://github.com/ShaikhAbdulZubair/NEXUS-OfflineMeshNetwork

Local Development Workspace: C:\Users\abdul\AndroidStudioProjects\NearMeshChat